

The first of these areas, he says, is the nature of the mind: “How do you model thought?” In AI research to date, he says, “what’s been missing is an ecology of models, a system that can solve problems in many ways”, as the mind does.

Part of this difficulty comes from the very nature of the human mind, evolved over billions of years as a complex mix of different functions and systems. “The pieces are very disparate; they’re not necessarily built in a compatible way,” Gershenfeld says. “There’s a similar pattern in AI research. There are lots of pieces that work well to solve some particular problem, and people have tried to fit everything into one of these.” Instead, he says, what’s needed are ways to “make systems made up of lots of pieces” that work together like the different elements of the mind. “Instead of searching for silver bullets, we’re looking at a range of models, trying to integrate them and aggregate them,” he says.

The second area of focus is memory. Much work in AI has tried to impose an artificial consistency of systems and rules on the messy, complex nature of human thought and memory. It’s now possible to accumulate the whole life experience of a person, and then reason using these data sets which are full of ambiguities and inconsistencies. That’s how the human brain functions – we don’t reason with precise truths. Computers need to learn ways to reason that work with, rather than avoid, ambiguity and inconsistency.

And the third focus of the new research has to do with what the MMP describes as “body”: “Computer science and physical science diverged decades ago,” Gershenfeld says. Computers are programmed by writing a sequence of lines of code, but the human mind doesn’t work that way. In the mind, everything happens everywhere all the time. A new approach to programming, called RALA (Reconfigurable Asynchronous Logic Automata) attempts to re-implement all of computer science on a base that looks like physics. The computations have physical units of time and space, so the description of the system aligns with the system it represents. This could lead to making computers that run with the fine-grained parallelism the brain uses.

Rethinking the Turing Test

Marvin Minsky, one of the pioneering researchers from AI’s early days and part of the MMP team, sees real hope for important contributions this time around. The inability of modern AI to pass the Turing Test is partly to blame on the way we assign credence to the human mind as a benchmark for intelligence, he says.